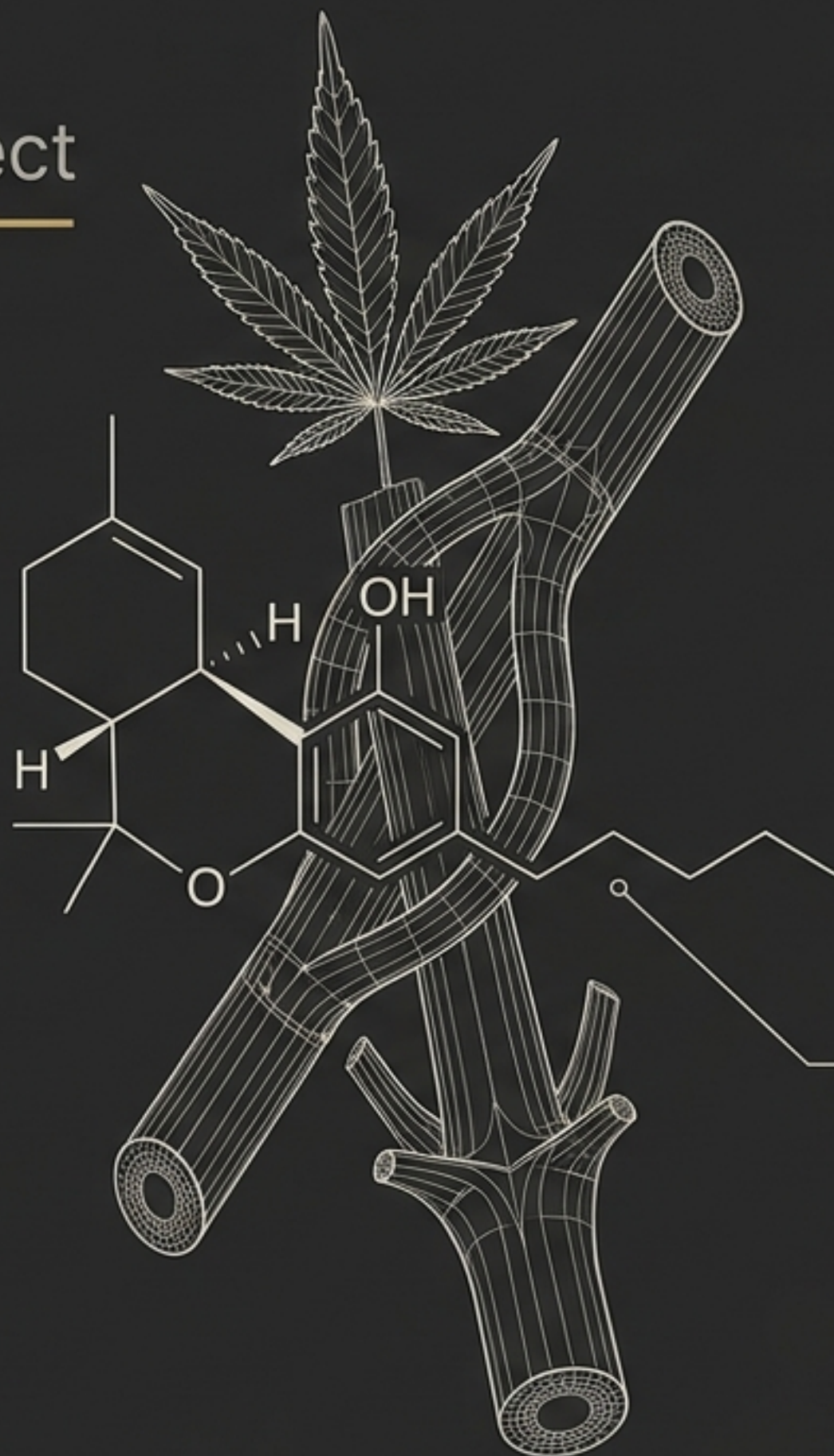


The Hempoxies

Manhattan Research Project

- Biomass Refining
- Structural Polymers
- Bioactive Medical
- Economic Ecosystem



Towards a 100% Hemp-Derived
Vitrimers Platform.

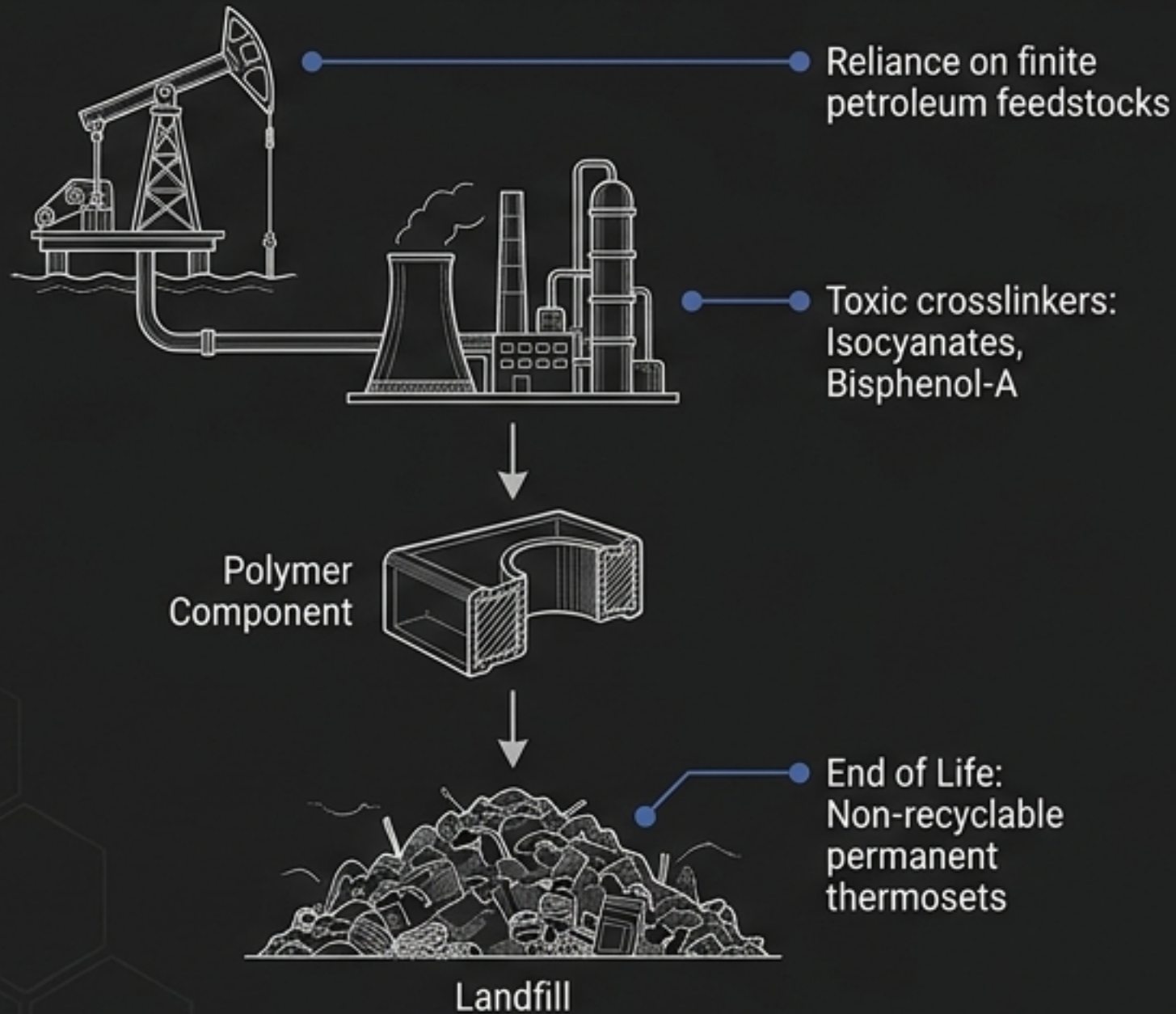
A complete biorefinery utilization framework for
Advanced Covalent Adaptable Networks (CANs).

Validated via ASTM/ISO protocols and published
as defensive prior art under CC BY 4.0.

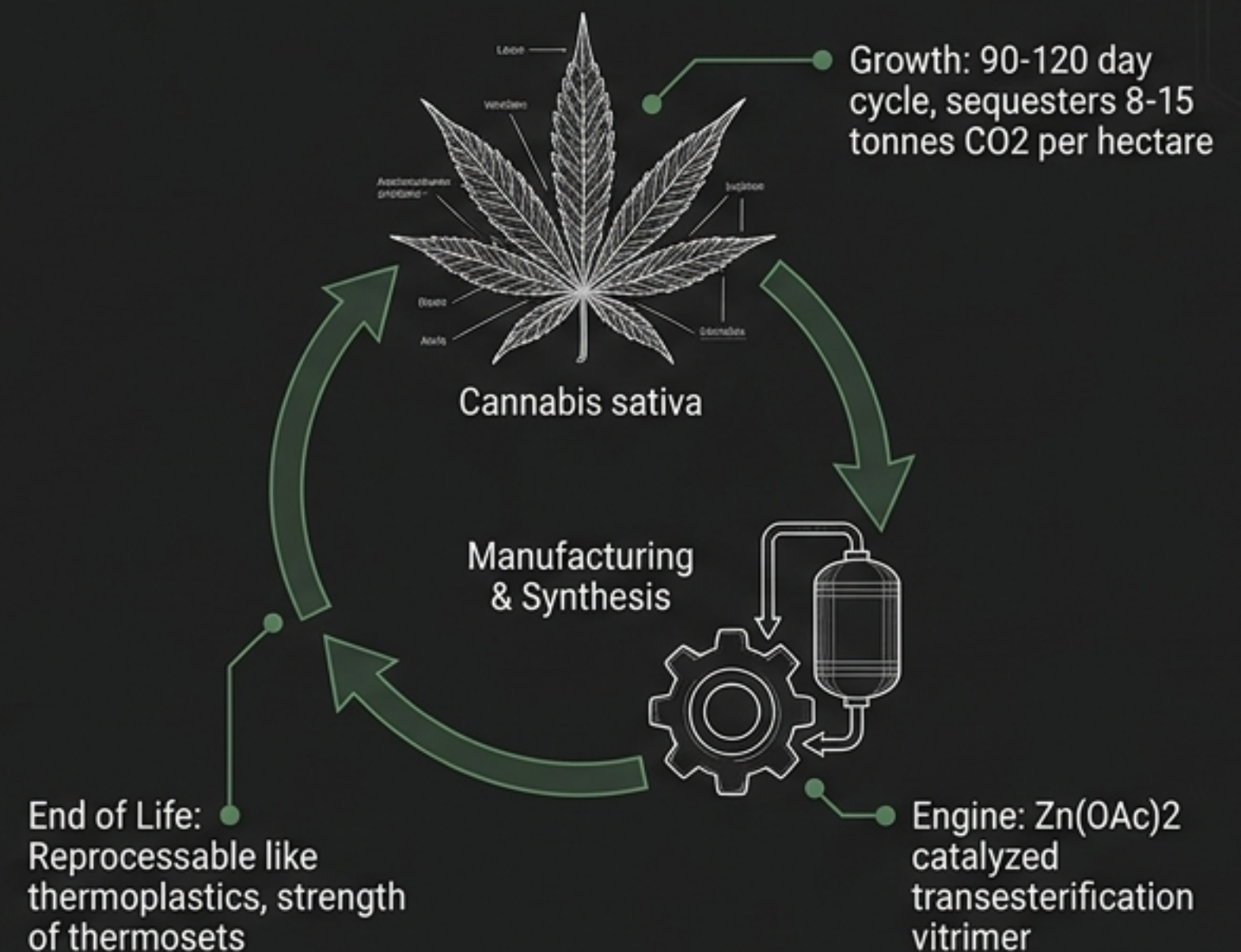
Target:
≥98%
Hemp-Derived
Organic Carbon
Content.

Replacing Petroleum Thermosets with a Regenerative Vitrimer Engine

Linear Petroleum Supply Chain

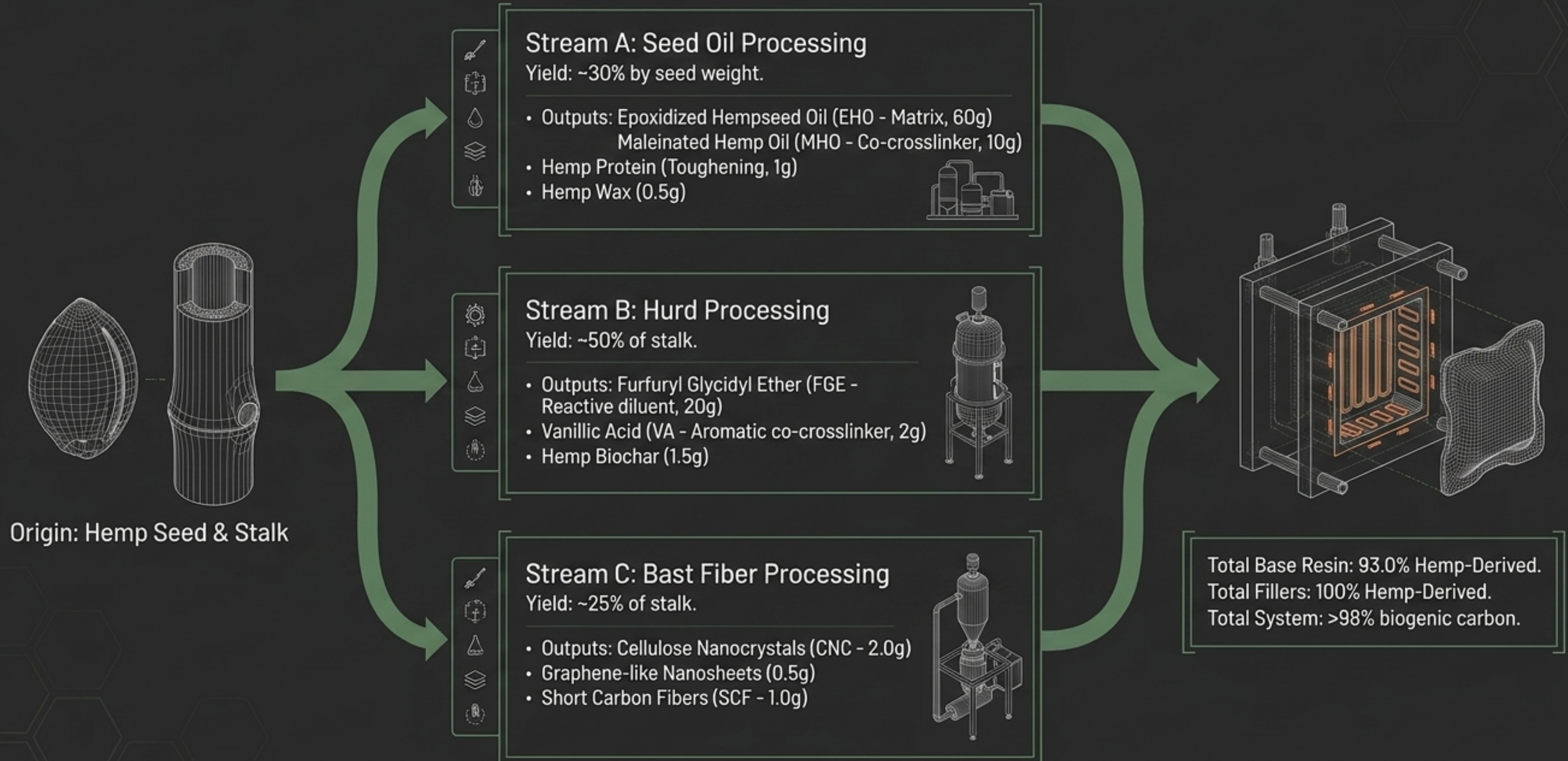


Circular Vitrimer Ecosystem



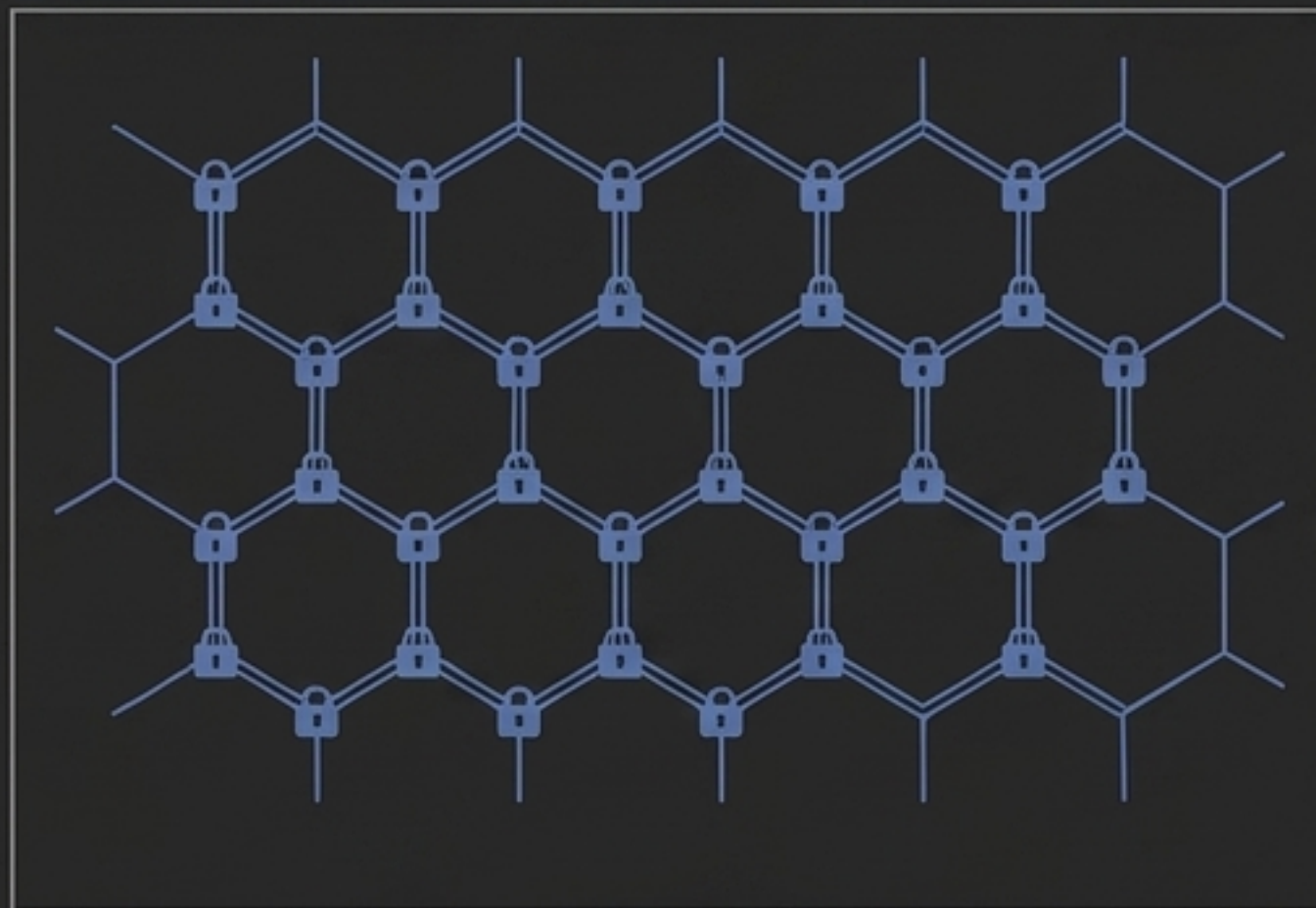
The goal is not merely to produce a composite, but to replace unsustainable plastics with the #1 biomass refinery plant ever discovered.

The Complete Biorefinery: Six Fractions into Seven Base Ingredients



The Vitrimer Mechanism: Dynamic Transesterification

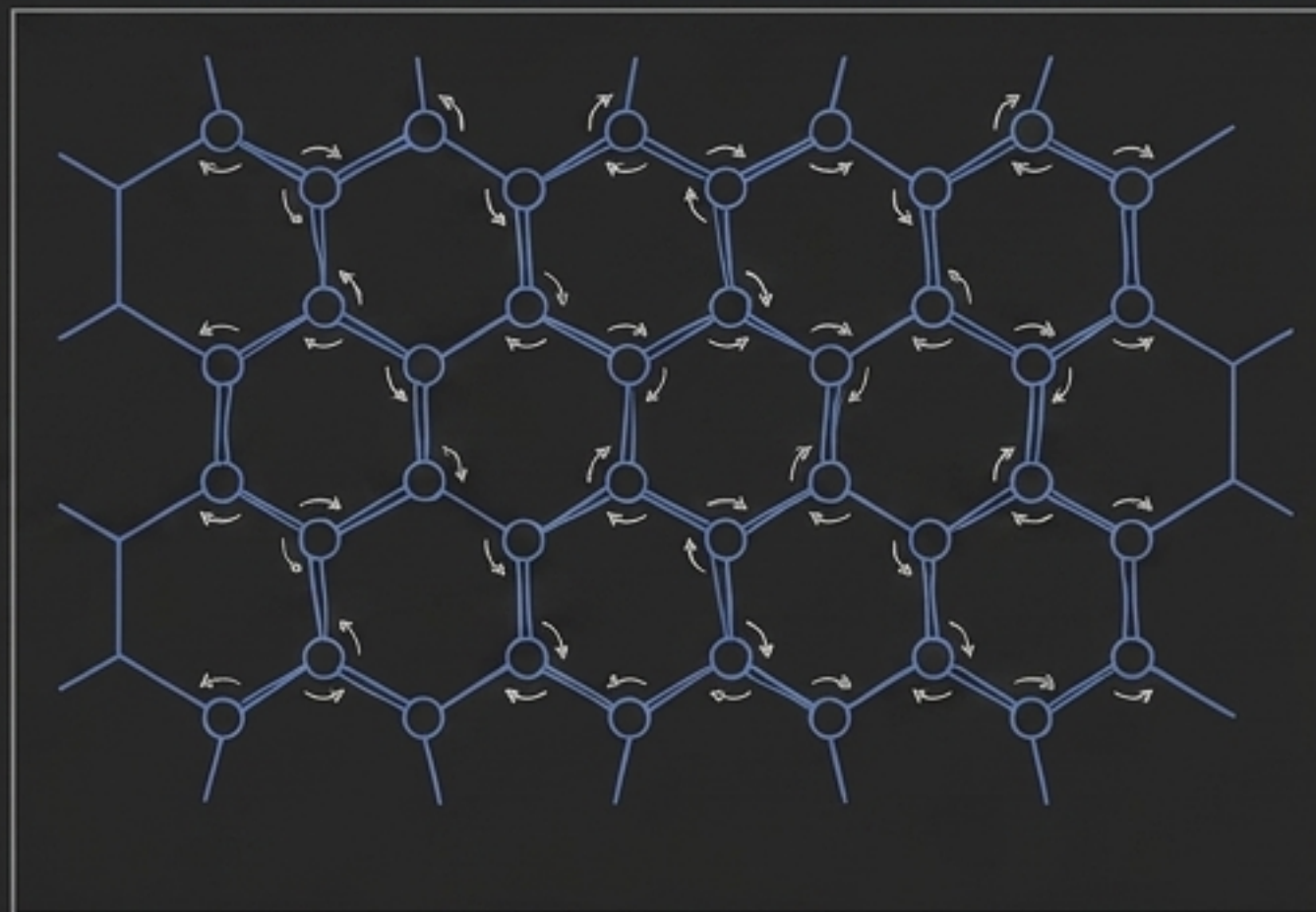
State 1: Service Temperature



Rigid lattice behaving like a high-strength thermoset ($T_g \geq 85^\circ\text{C}$).

+ Heat / Catalyst

State 2: Reprocessing at 180°C



Thermal activation enables ester linkages to exchange, allowing flow without losing structural integrity.

Catalyst:
5 mol% Zinc Acetate
Dihydrate ($\text{Zn}(\text{OAc})_2$)

Kinetics Target:
Stress relaxation
 $\tau^* < 100 \text{ s}$ at 160°C

Activation Energy:
 $E_a \approx 80\text{-}120 \text{ kJ/mol}$

Maxwell Model:
 $E(t) = E_0 \exp(-t/\tau^*)$

Hierarchical Reinforcement Across Five Spatial Scales

Panel 1: Macro-Scale (mm)

3-6 mm Short Hemp
Carbon Fibers (SCF)

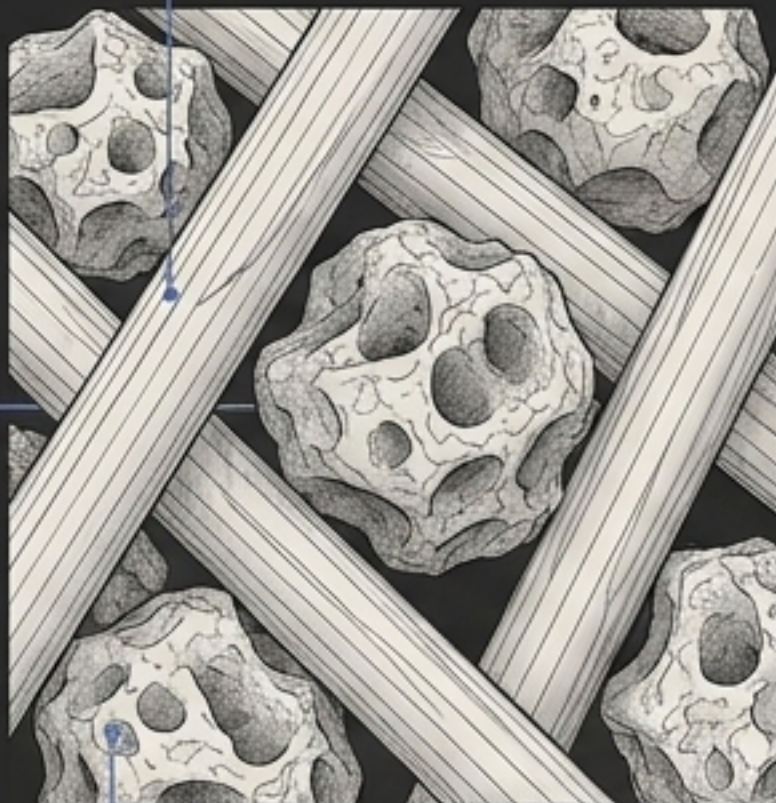


3-6 mm Short Hemp
Carbon Fibers (SCF)

Function: Primary structural load-bearing reinforcement (>1 GPa single-fiber strength).

Panel 2: Micro-Scale (μm)

50-100 μm Hemp Biochar

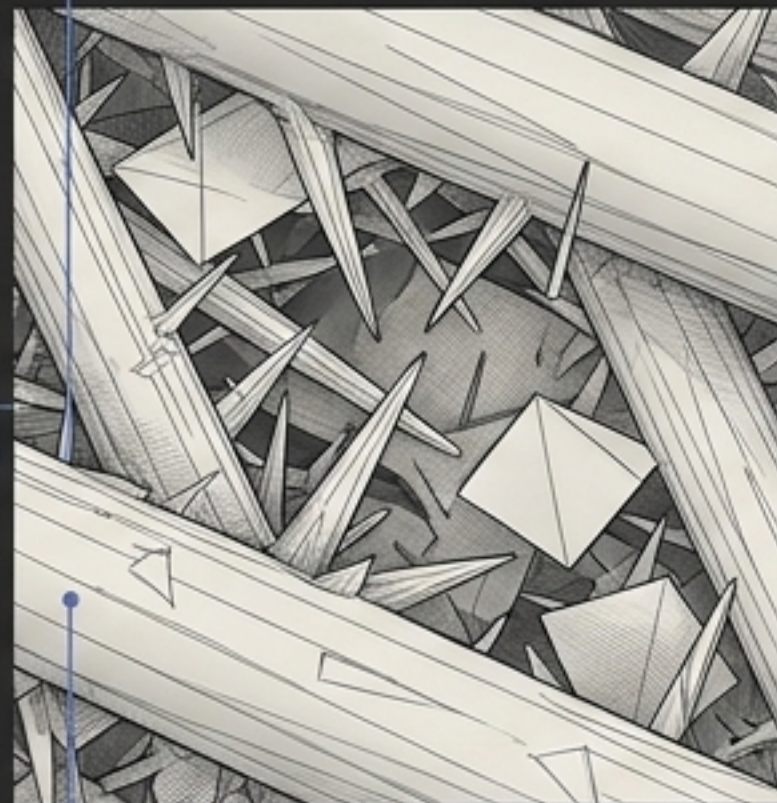


50-100 μm Hemp Biochar

Function: Interstitial micro-filler providing compressive support (Fixed Carbon >70 wt%).

Panel 3: Nano-Scale (nm)

Cellulose Nanocrystals (CNC) &
Graphene Nanosheets (2-10 nm)

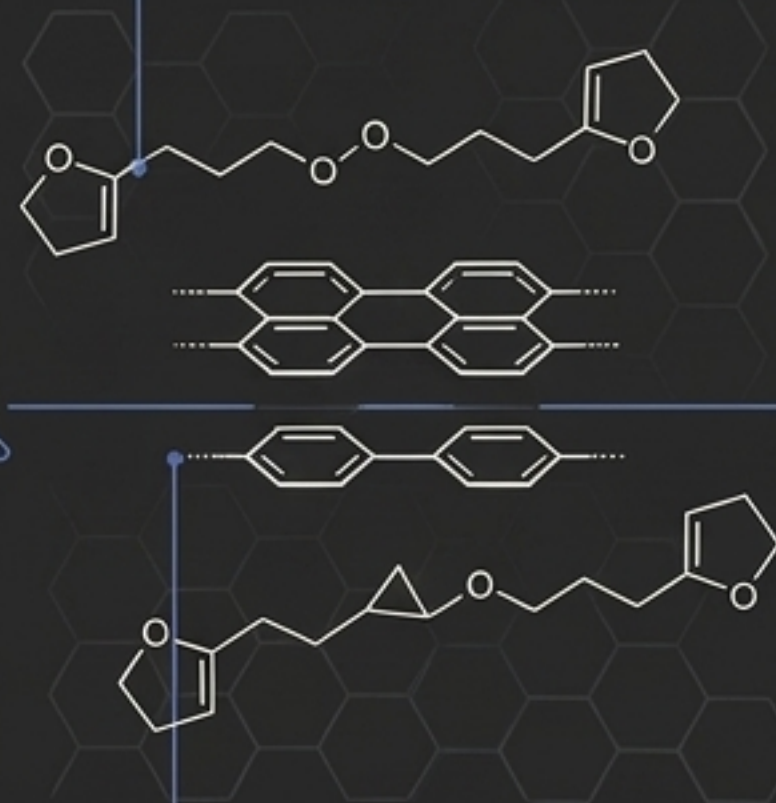


Cellulose Nanocrystals (CNC) &
Graphene Nanosheets (2-10 nm)

Function: Crack pinning, high-aspect ratio stiffness, and electrical/thermal percolation.

Panel 4: Molecular Scale (0.1 nm)

Furfuryl Glycidyl Ether
(FGE) Molecules



Furfuryl Glycidyl Ether
(FGE) Molecules

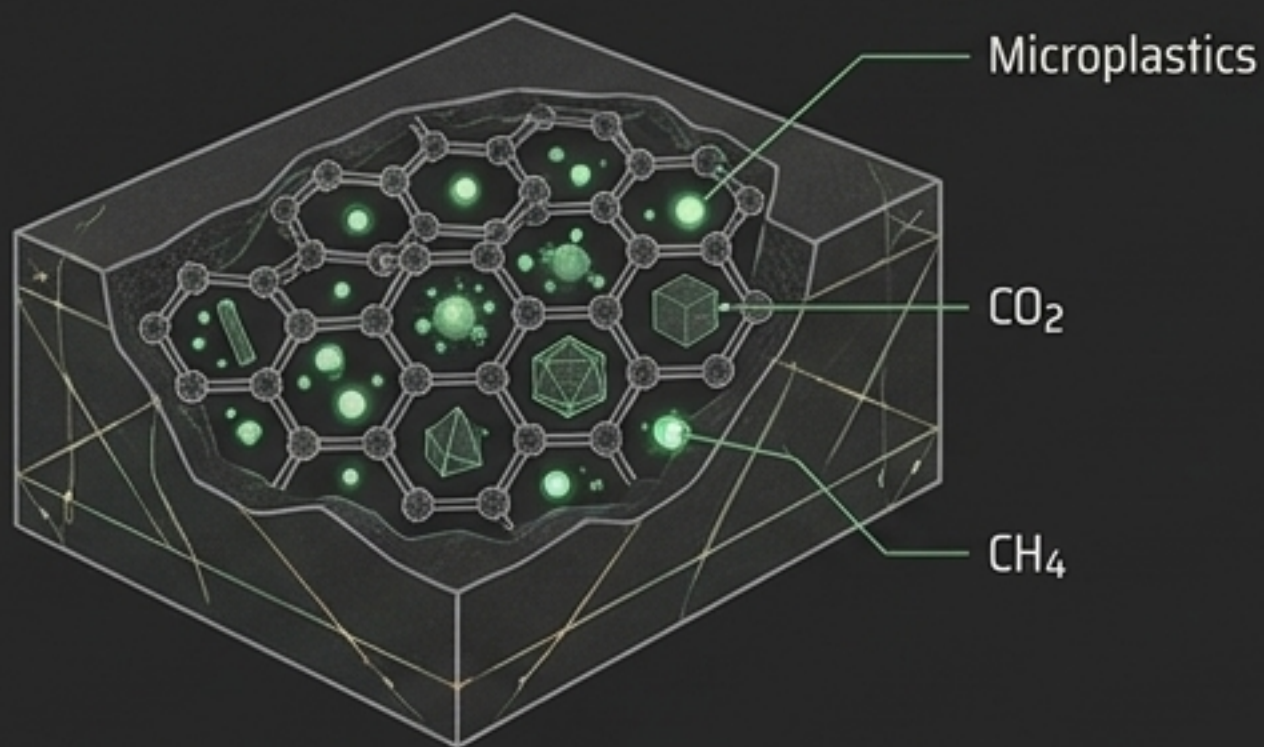
Function: π - π stacking for interfacial coupling and resin viscosity reduction.

The Hempoxies Industrial Variant Matrix

Variant	Target Incumbent	Key Formulation Tweaks	Target Metrics
HX-S200 (Structural)	Aluminum 6061 / Mild Steel	55 wt% MHO, 25 phr aligned short carbon fiber.	Specific Modulus 28.1 GPa·cm ³ /g (+9.8% vs. Al), Tensile >45 MPa.
HX-C300 (Conductive)	Nickel-Coated ABS Plastic	High loading of Hemp Graphene-like Nanosheets.	EMI Shielding >40 dB @ 1 GHz, Volume Resistivity <100 Ω·cm.
HX-L400 (Lightweight Foam)	Extruded Polystyrene (XPS)	Supercritical fluid foaming.	Thermal Conductivity <0.040 W/m-K.
HX-G100 (General Purpose)	Petroleum DGEBA Epoxy	Balanced MHO/CA (25 eq.% MHO).	Tg 60-75°C, Modulus 1.8-2.2 GPa.

De-risking the “Trojan Horse”: HX-R500 Remediation Variant

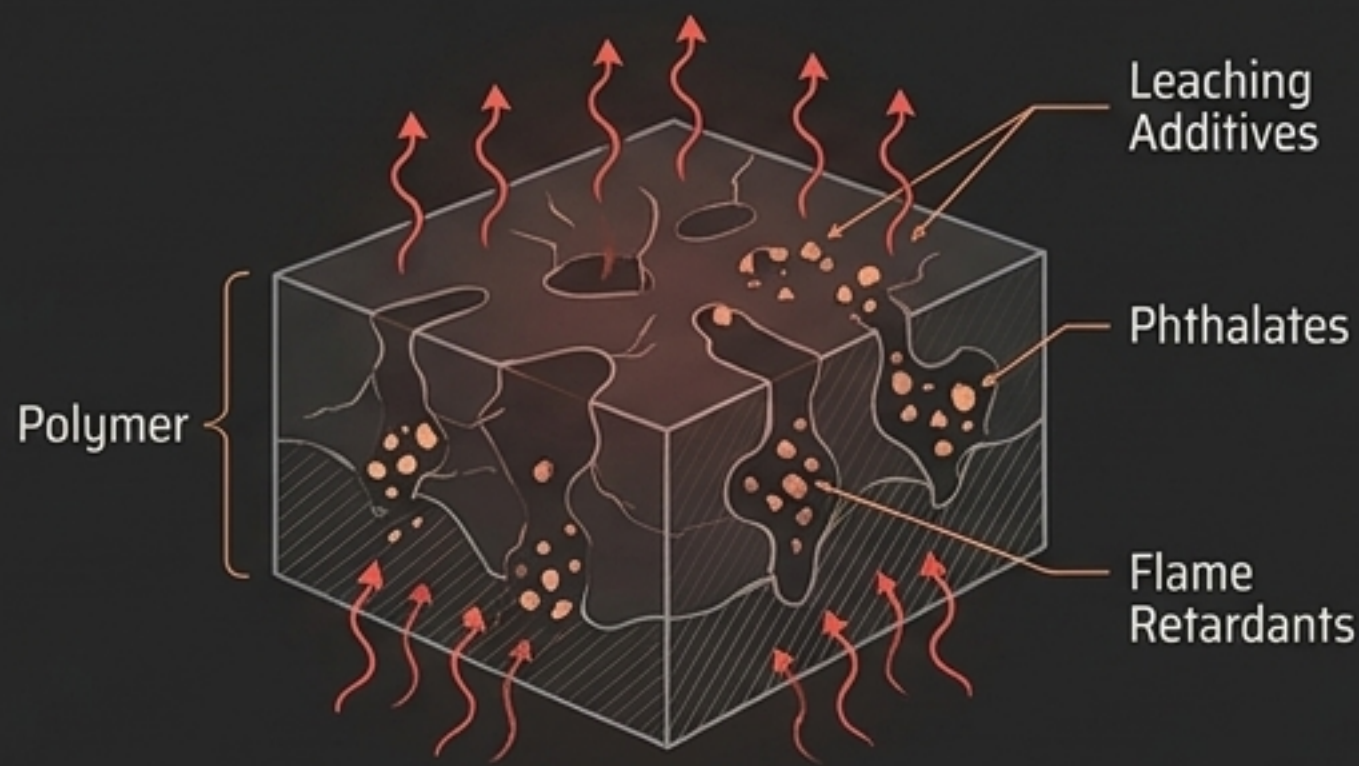
THE REWARD



Concept: Sequestering harvested microplastics and captured CO₂/CH₄ into a durable, load-bearing infrastructure composite.

Impact: Actively removing existing waste streams while displacing concrete.

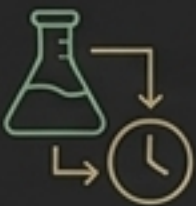
THE RISK



The 'Trojan Horse' Threat: Microplastics are not inert; they contain toxic additives (phthalates, brominated flame retardants).

Vulnerability: Vitrimer thermal cycling (180°C) could accelerate additive leaching into the environment, destroying ESG credibility.

THE MITIGATION PROTOCOL



Action: Non-negotiable, multi-year LC-MS leaching study.



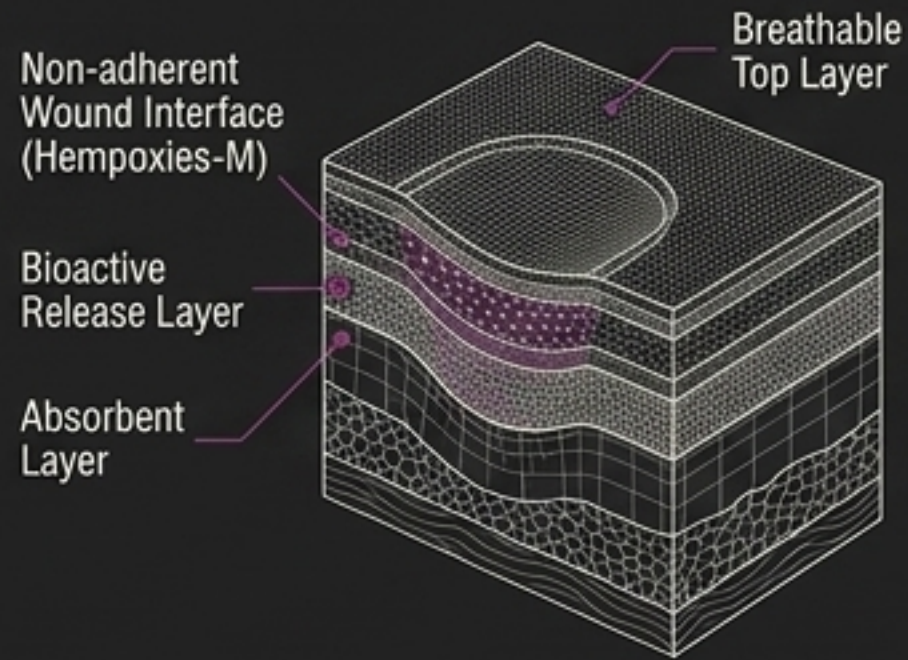
Testing Matrix: ISO/ASTM protocols simulating UV exposure, fresh/salt water immersion, and prolonged thermal cycling to quantify release rates.



Target: Leaching < Detection Limit.

Frontier Applications: Hempoxies-M Bioactive Medical Composites

Antimicrobial Wound Dressing

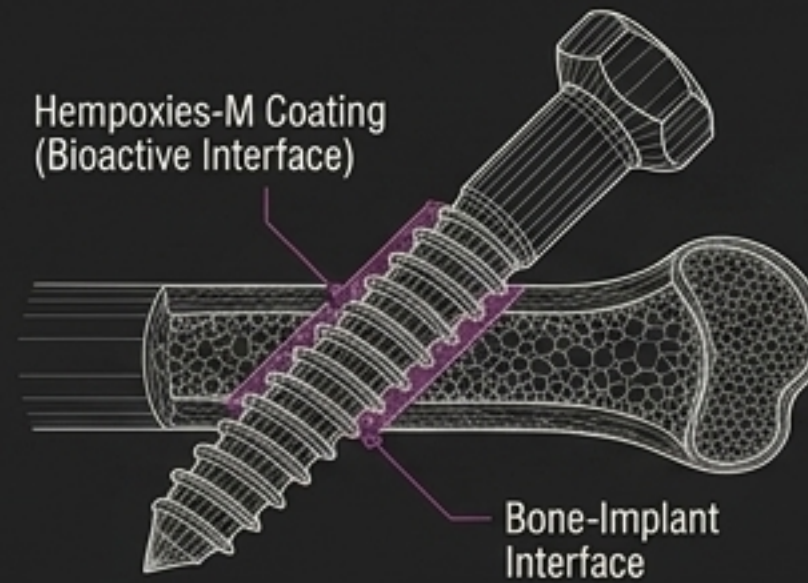


Bioactives: β -Caryophyllene, Myrcene, Cannflavin A/B.

Mechanism: Sustained release into wound exudate. Broad-spectrum antimicrobial (*S. aureus*, *E. coli*) via membrane disruption.

Release Target: 60-70% over 7 days (Korsmeyer-Peppas $n = 0.50-0.65$).

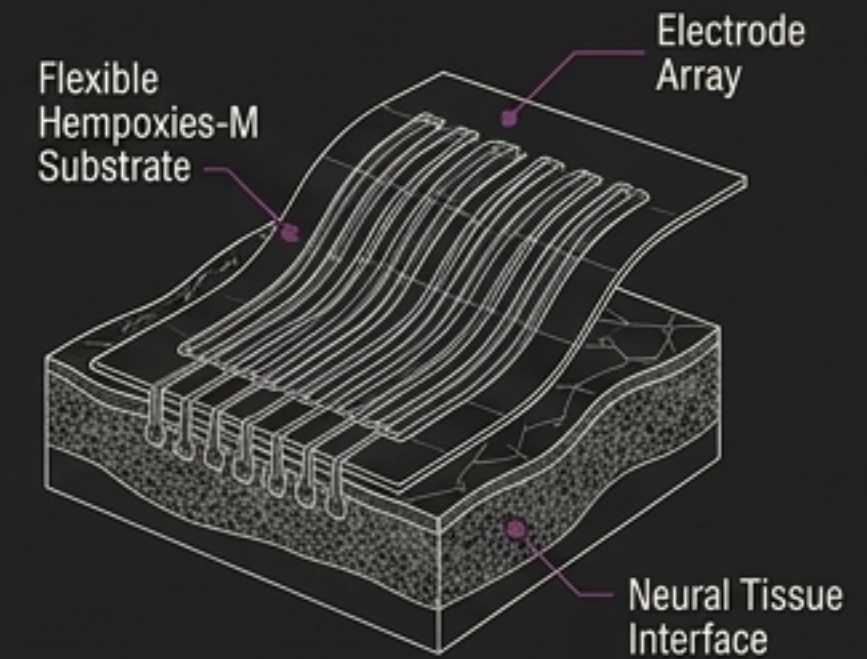
Orthopedic Implant Coating



Bioactives: CBD, CBG.

Mechanism: Localized anti-inflammatory release at the bone-implant interface to reduce osteolysis and improve osseointegration (Inhibits COX-2, TNF- α).

Neural Interface Device



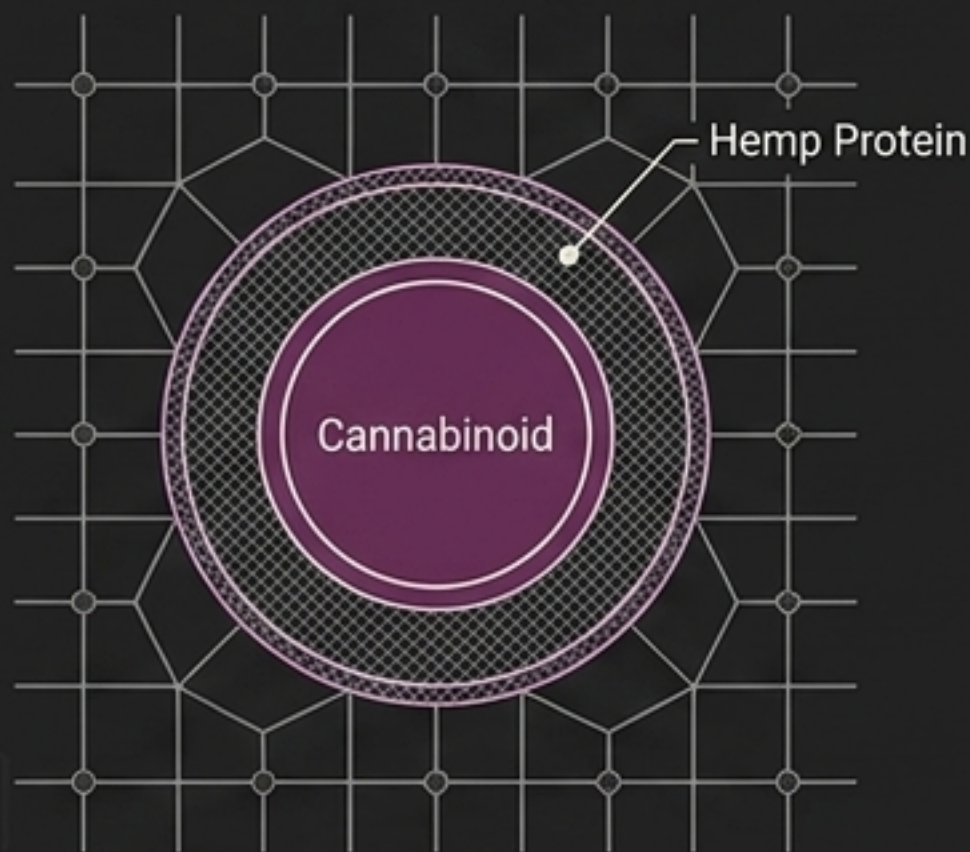
Bioactives: CBG, Hempoxies matrix flexibility.

Mechanism: Neuroprotective release reduces glial scarring; soft vitrimer matches neural tissue modulus better than rigid epoxies.

Surviving the Cure: Three Cannabinoid Thermal Protection Strategies

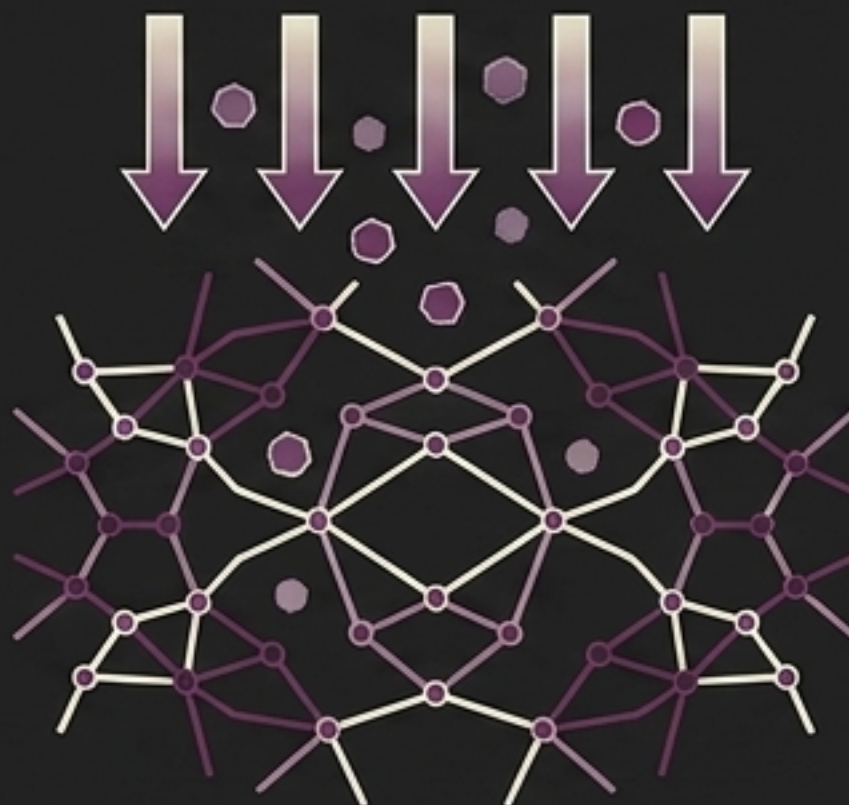
The Problem: $\Delta 9$ -THC degrades at 120°C, CBD degrades at 140°C. The Hempoxies vitrimer requires a 110°C/2h + 140°C/4h thermal cure.

Strategy A: Core-Shell Nanoparticles



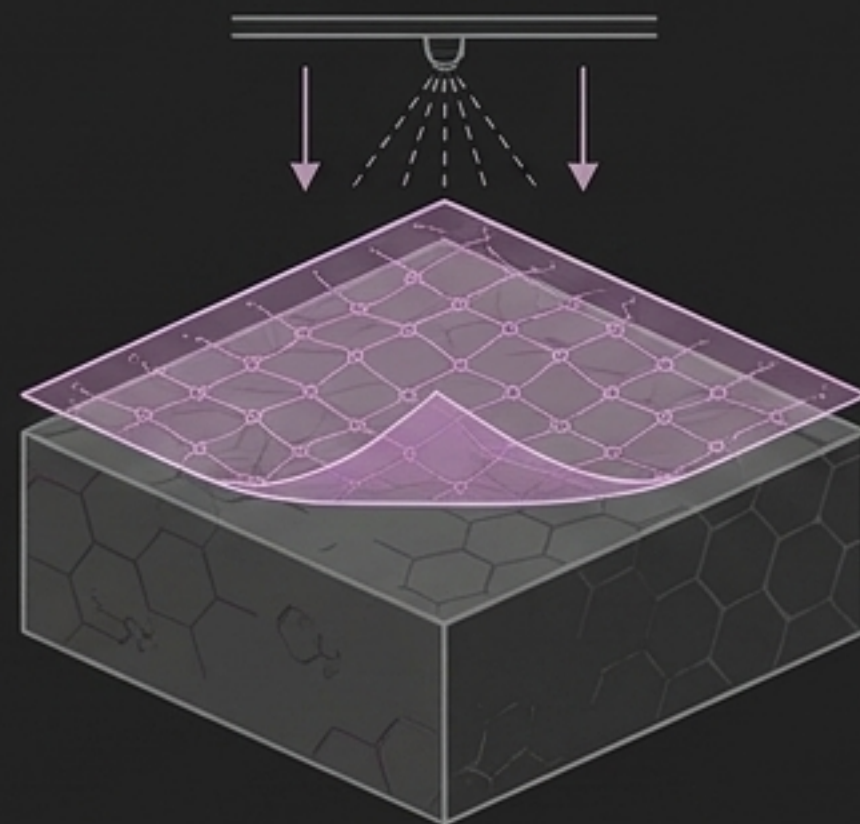
Method: Microencapsulation (150-400 nm) before curing. Shell degrades above 180°C, protecting the core during the 140°C cure.

Strategy B: Matrix Swelling / Infusion



Method: Post-cure infusion. Vitrimer is heated to 160°C for 15 mins to relax bonds ($\tau^* < 100$ s). Composite swells, absorbing CBD solution, and locks upon cooling.

Strategy C: Surface Laminate



Method: Zero thermal stress. Cannabinoid-loaded hemp protein film is spray-coated onto the cured surface and cross-linked with transglutaminase at room temperature.

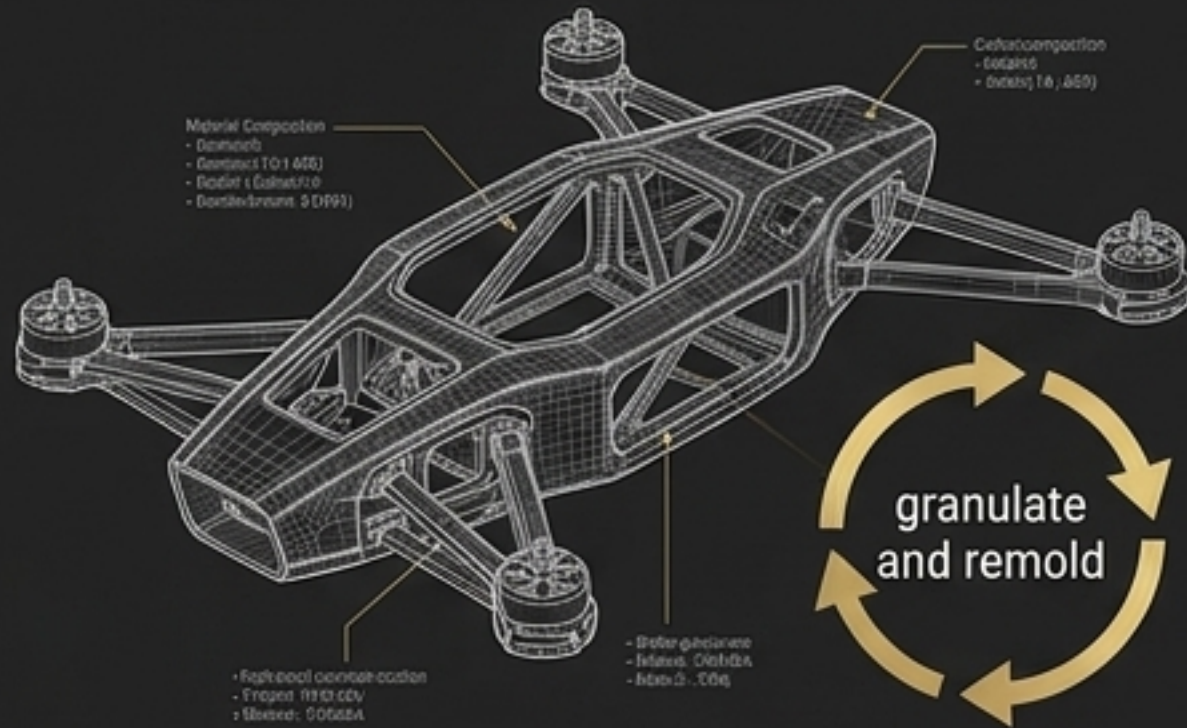
The Medical Regulatory Pathway Matrix

Formulation	Health Claim	FDA Status (US)	Health Canada Status (CAN)
Hemp Seed Oil + Carbon Fillers	None (Structural only).	GRAS / Standard Class II medical device (510(k)).	Class I-II device.
Terpene-Flavonoid Module (Module A)	Antimicrobial / Anti-inflammatory.	Cosmetic / Topical.	Natural Health Product (NHP) if <10 ppm THC.
Cannabinoid Module (Module B)	Therapeutic claims (pain, inflammation).	Prescription drug (NDA/BLA) / Drug-Device Combination.	Prescription drug / Dual-licensed device (DIN required).

Launch Module A (Terpenes) immediately via NHP/GRAS pathways for revenue generation, while partnering with pharma for Module B (CBD/CBG) IND-enabling studies.

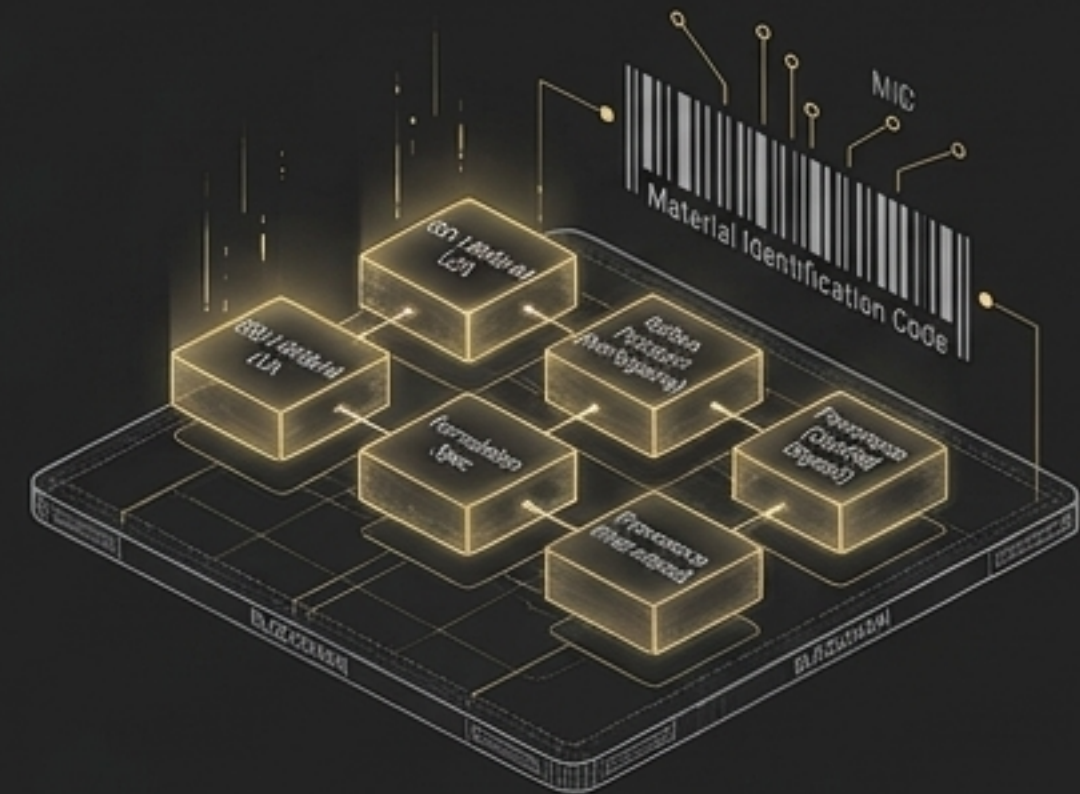
The Dual-Use Economic Model: Physical Utility Meets Digital ESG Asset

The Physical Asset



The physical composite acts as a carbon-negative industrial feedstock, replacing aluminum and ABS with superior specific strength.

The Digital Twin



The physical asset is tied to an immutable digital passport. It tracks ISO 14040/44 verified LCA (net-negative CO2eq), precise formulation, and provenance from certified organic farms.

Hempoxies is not just a plastic replacement. It is a traceable, fungible, carbon-negative currency designed for the modern ESG investing landscape.

The Execution Roadmap: The Atlantic Canada Innovation Engine



Open-Source Framework & Defensive Prior Art

Collaboration Tiers



Tier 1 (Verification)
Replicate base EHO/CA vitrimer.



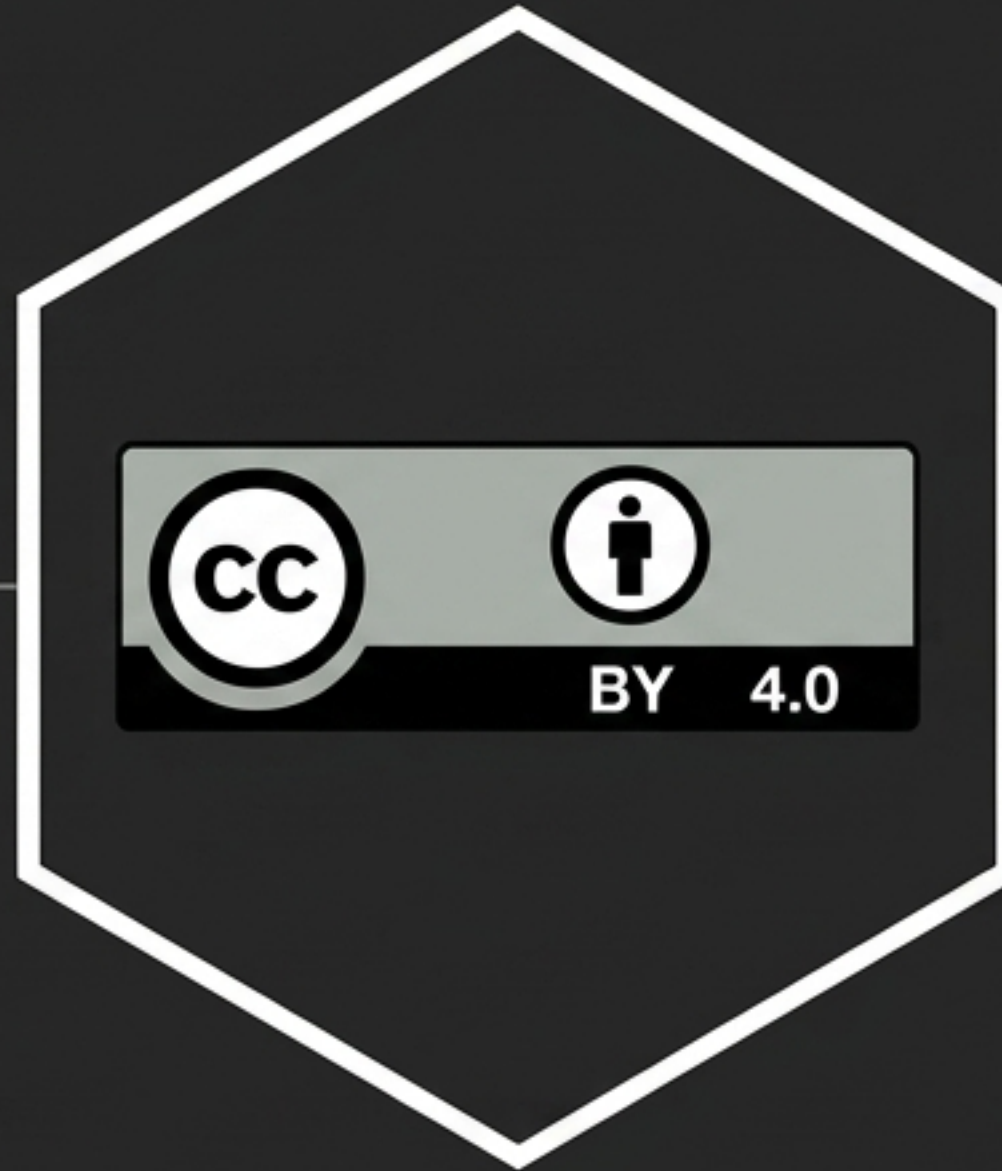
Tier 2 (Innovation)
Improve CNC extraction or lignin oxidation.



Tier 3 (Application)
Fabricate structural or medical prototypes.



Tier 4 (Scale-Up)
Pilot production >100 kg/batch.



IP Strategy



All formulations, protocols, and raw data are published under Creative Commons (CC BY 4.0) within 30 days of validation.

This ensures the “Complete Hemp” formulation remains a public good, actively preventing monopolization of the biorefinery platform.

This is a biorefinery operating system and a regenerative materials platform rooted in the most versatile biomass organism on Earth. Share freely. Attribute always.